In[ ]:=
```
EYoung1 = 180 * 10^9;
EYoung2 = 180 * 10^9;
b = 5 * 10^-3;
h1 = 0.5 * 10^-3;
h2 = 0.5 * 10^-3;
EI1Val = EYoung1 * b * h1^3 / 12;
EI2Val = EYoung2 * b * h2^3 / 12;
l1Val = 0.075;
l2Val = 0.07;
ρ1Val = 7800 * (b * h1);
ρ2Val = 7800 * (b * h2);
MVal = 0.236;
m1Val = 0.05;
m2Val = 0.04;
rad = 0.01;
lm1 = m1Val / (7800 * Pi * rad^2);
J1Val = m1Val * (lm1^2 / 3 + rad^2 / 4);
lm2 = m2Val / (7800 * Pi * rad^2);
J2Val = m2Val * (lm2^2 / 3 + rad^2 / 4);
```

In[ ]:=
```
(*Define symbolic variables*)

ClearAll[s, β1, β2, A1i, B1i, C1i, D1i,
  A2i, B2i, C2i, D2i, EI1, EI2, l1, l2, J, ρ1, ρ2, m1, m2, t];

ω21 = ((β1^4) * EI1 / ρ1) ^0.5;
ω22 = ((β2^4) * EI2 / ρ2) ^0.5;

(*Define the mode shape equation*)
ϕ1 = A1i * Sin[β1 * s] + B1i * Cos[β1 * s] + C1i * Sinh[β1 * s] + D1i * Cosh[β1 * s];
ϕ2 = A2i * Sin[β2 * s] + B2i * Cos[β2 * s] + C2i * Sinh[β2 * s] + D2i * Cosh[β2 * s];

(*Derivatives with respect to s*)
ϕ1d1 = D[ϕ1, s];
ϕ1d2 = D[ϕ1, {s, 2}];
ϕ1d3 = D[ϕ1, {s, 3}];

ϕ2d1 = D[ϕ2, s];
ϕ2d2 = D[ϕ2, {s, 2}];
ϕ2d3 = D[ϕ2, {s, 3}];

(*Boundary conditions*)
bc11 = (ϕ1 /. s -> 0) == 0;
bc12 = (ϕ1d1 /. s -> 0) == 0;
bc13 = EI1 * (ϕ1d2 /. s -> l1) - ω21^2 * J1 * (ϕ1d1 /. s -> l1) == 0;
bc14 = EI1 * (ϕ1d3 /. s -> l1) + ω21^2 * m1 * (ϕ1 /. s -> l1) == 0;

bc21 = (ϕ2 /. s -> 0) == 0;
bc22 = (ϕ2d1 /. s -> 0) == 0;
bc23 = EI2 * (ϕ2d2 /. s -> l2) - ω22^2 * J2 * (ϕ2d1 /. s -> l2) == 0;
bc24 = EI2 * (ϕ2d3 /. s -> l2) + ω22^2 * m2 * (ϕ2 /. s -> l2) == 0;

(*Collect terms involving A,B,C,and D*)
bc11Coll = Collect[bc11, {A1i, B1i, C1i, D1i}]
bc12Coll = Collect[bc12, {A1i, B1i, C1i, D1i}]
bc13Coll = Collect[bc13, {A1i, B1i, C1i, D1i}]
bc14Coll = Collect[bc14, {A1i, B1i, C1i, D1i}]

bc21Coll = Collect[bc21, {A2i, B2i, C2i, D2i}]
bc22Coll = Collect[bc22, {A2i, B2i, C2i, D2i}]
bc23Coll = Collect[bc23, {A2i, B2i, C2i, D2i}]
bc24Coll = Collect[bc24, {A2i, B2i, C2i, D2i}]
```

Out[ ]= B1i + D1i == 0

Out[ ]= A1i β1 + C1i β1 == 0

Out[•]= $$A1i\left(-J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Cos}[l1\,\beta1]-EI1\,\beta1^2\,\mathrm{Sin}[l1\,\beta1]\right)+B1i\left(-EI1\,\beta1^2\,\mathrm{Cos}[l1\,\beta1]+J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Sin}[l1\,\beta1]\right)+C1i\left(-J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Cosh}[l1\,\beta1]+EI1\,\beta1^2\,\mathrm{Sinh}[l1\,\beta1]\right)+D1i\left(EI1\,\beta1^2\,\mathrm{Cosh}[l1\,\beta1]-J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Sinh}[l1\,\beta1]\right)==0$$

Out[•]= $$B1i\left(m1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Cos}[l1\,\beta1]+EI1\,\beta1^3\,\mathrm{Sin}[l1\,\beta1]\right)+A1i\left(-EI1\,\beta1^3\,\mathrm{Cos}[l1\,\beta1]+m1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Sin}[l1\,\beta1]\right)+D1i\left(m1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Cosh}[l1\,\beta1]+EI1\,\beta1^3\,\mathrm{Sinh}[l1\,\beta1]\right)+C1i\left(EI1\,\beta1^3\,\mathrm{Cosh}[l1\,\beta1]+m1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Sinh}[l1\,\beta1]\right)==0$$

Out[•]= $$B2i+D2i==0$$

Out[•]= $$A2i\,\beta2+C2i\,\beta2==0$$

Out[•]= $$A2i\left(-J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Cos}[l2\,\beta2]-EI2\,\beta2^2\,\mathrm{Sin}[l2\,\beta2]\right)+B2i\left(-EI2\,\beta2^2\,\mathrm{Cos}[l2\,\beta2]+J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Sin}[l2\,\beta2]\right)+C2i\left(-J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Cosh}[l2\,\beta2]+EI2\,\beta2^2\,\mathrm{Sinh}[l2\,\beta2]\right)+D2i\left(EI2\,\beta2^2\,\mathrm{Cosh}[l2\,\beta2]-J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Sinh}[l2\,\beta2]\right)==0$$

Out[•]= $$B2i\left(m2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Cos}[l2\,\beta2]+EI2\,\beta2^3\,\mathrm{Sin}[l2\,\beta2]\right)+A2i\left(-EI2\,\beta2^3\,\mathrm{Cos}[l2\,\beta2]+m2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Sin}[l2\,\beta2]\right)+D2i\left(m2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Cosh}[l2\,\beta2]+EI2\,\beta2^3\,\mathrm{Sinh}[l2\,\beta2]\right)+C2i\left(EI2\,\beta2^3\,\mathrm{Cosh}[l2\,\beta2]+m2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Sinh}[l2\,\beta2]\right)==0$$

In[ ]:=
```
(*Define the boundary conditions (already collected)*)
boundaryConditionsbeam1 = {bc11Coll, bc12Coll, bc13Coll, bc14Coll};
boundaryConditionsbeam2 = {bc21Coll, bc22Coll, bc23Coll, bc24Coll};

(*Extract the coefficient matrix*)
coefficientMatrixbeam1 =
  CoefficientArrays[boundaryConditionsbeam1, {A1i, B1i, C1i, D1i}][[2]];
coefficientMatrixbeam2 =
  CoefficientArrays[boundaryConditionsbeam2, {A2i, B2i, C2i, D2i}][[2]];

(*Display the coefficient matrix*)
MatrixForm[coefficientMatrixbeam1]
MatrixForm[coefficientMatrixbeam2]
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} 0 & 1. \\ \beta 1 & 0 \\ -1.\ J1\,\beta 1\left(\frac{EI1\,\beta 1^4}{\rho 1}\right)^{1.}\mathrm{Cos}[l1\,\beta 1] - 1.\ EI1\,\beta 1^2\,\mathrm{Sin}[l1\,\beta 1] & -1.\ EI1\,\beta 1^2\,\mathrm{Cos}[l1\,\beta 1] + J1\,\beta 1\left(\frac{EI1\,\beta 1^4}{\rho 1}\right)^{1.}\mathrm{Sin} \\ -1.\ EI1\,\beta 1^3\,\mathrm{Cos}[l1\,\beta 1] + m1\left(\frac{EI1\,\beta 1^4}{\rho 1}\right)^{1.}\mathrm{Sin}[l1\,\beta 1] & m1\left(\frac{EI1\,\beta 1^4}{\rho 1}\right)^{1.}\mathrm{Cos}[l1\,\beta 1] + EI1\,\beta 1^3\,\mathrm{Sin}[l1 \end{pmatrix}$$

Out[ ]//MatrixForm=

$$\begin{pmatrix} 0 & 1. \\ \beta 2 & 0 \\ -1.\ J2\,\beta 2\left(\frac{EI2\,\beta 2^4}{\rho 2}\right)^{1.}\mathrm{Cos}[l2\,\beta 2] - 1.\ EI2\,\beta 2^2\,\mathrm{Sin}[l2\,\beta 2] & -1.\ EI2\,\beta 2^2\,\mathrm{Cos}[l2\,\beta 2] + J2\,\beta 2\left(\frac{EI2\,\beta 2^4}{\rho 2}\right)^{1.}\mathrm{Sin} \\ -1.\ EI2\,\beta 2^3\,\mathrm{Cos}[l2\,\beta 2] + m2\left(\frac{EI2\,\beta 2^4}{\rho 2}\right)^{1.}\mathrm{Sin}[l2\,\beta 2] & m2\left(\frac{EI2\,\beta 2^4}{\rho 2}\right)^{1.}\mathrm{Cos}[l2\,\beta 2] + EI2\,\beta 2^3\,\mathrm{Sin}[l2 \end{pmatrix}$$

In[∘]:=
```
(*Calculate the determinant of the coefficient matrix*)
determinantbeam1 = Det[coefficientMatrixbeam1]
determinantbeam2 = Det[coefficientMatrixbeam2]
```

Out[∘]=

$$0. + 1.\ EI1^2\ \beta1^6\ \mathrm{Cos}[l1\ \beta1]^2 + 1.\ J1\ m1\ \beta1^2 \left(\frac{EI1\ \beta1^4}{\rho1}\right)^{2.} \mathrm{Cos}[l1\ \beta1]^2 +$$
$$2.\ EI1^2\ \beta1^6\ \mathrm{Cos}[l1\ \beta1]\ \mathrm{Cosh}[l1\ \beta1] - 2.\ J1\ m1\ \beta1^2 \left(\frac{EI1\ \beta1^4}{\rho1}\right)^{2.} \mathrm{Cos}[l1\ \beta1]\ \mathrm{Cosh}[l1\ \beta1] +$$
$$1.\ EI1^2\ \beta1^6\ \mathrm{Cosh}[l1\ \beta1]^2 + 1.\ J1\ m1\ \beta1^2 \left(\frac{EI1\ \beta1^4}{\rho1}\right)^{2.} \mathrm{Cosh}[l1\ \beta1]^2 -$$
$$2.\ EI1\ m1\ \beta1^3 \left(\frac{EI1\ \beta1^4}{\rho1}\right)^{1.} \mathrm{Cosh}[l1\ \beta1]\ \mathrm{Sin}[l1\ \beta1] -$$
$$2.\ EI1\ J1\ \beta1^5 \left(\frac{EI1\ \beta1^4}{\rho1}\right)^{1.} \mathrm{Cosh}[l1\ \beta1]\ \mathrm{Sin}[l1\ \beta1] + 1.\ EI1^2\ \beta1^6\ \mathrm{Sin}[l1\ \beta1]^2 +$$
$$1.\ J1\ m1\ \beta1^2 \left(\frac{EI1\ \beta1^4}{\rho1}\right)^{2.} \mathrm{Sin}[l1\ \beta1]^2 + 2.\ EI1\ m1\ \beta1^3 \left(\frac{EI1\ \beta1^4}{\rho1}\right)^{1.} \mathrm{Cos}[l1\ \beta1]\ \mathrm{Sinh}[l1\ \beta1] -$$
$$2.\ EI1\ J1\ \beta1^5 \left(\frac{EI1\ \beta1^4}{\rho1}\right)^{1.} \mathrm{Cos}[l1\ \beta1]\ \mathrm{Sinh}[l1\ \beta1] -$$
$$1.\ EI1^2\ \beta1^6\ \mathrm{Sinh}[l1\ \beta1]^2 - 1.\ J1\ m1\ \beta1^2 \left(\frac{EI1\ \beta1^4}{\rho1}\right)^{2.} \mathrm{Sinh}[l1\ \beta1]^2$$

Out[∘]=

$$0. + 1.\ EI2^2\ \beta2^6\ \mathrm{Cos}[l2\ \beta2]^2 + 1.\ J2\ m2\ \beta2^2 \left(\frac{EI2\ \beta2^4}{\rho2}\right)^{2.} \mathrm{Cos}[l2\ \beta2]^2 +$$
$$2.\ EI2^2\ \beta2^6\ \mathrm{Cos}[l2\ \beta2]\ \mathrm{Cosh}[l2\ \beta2] - 2.\ J2\ m2\ \beta2^2 \left(\frac{EI2\ \beta2^4}{\rho2}\right)^{2.} \mathrm{Cos}[l2\ \beta2]\ \mathrm{Cosh}[l2\ \beta2] +$$
$$1.\ EI2^2\ \beta2^6\ \mathrm{Cosh}[l2\ \beta2]^2 + 1.\ J2\ m2\ \beta2^2 \left(\frac{EI2\ \beta2^4}{\rho2}\right)^{2.} \mathrm{Cosh}[l2\ \beta2]^2 -$$
$$2.\ EI2\ m2\ \beta2^3 \left(\frac{EI2\ \beta2^4}{\rho2}\right)^{1.} \mathrm{Cosh}[l2\ \beta2]\ \mathrm{Sin}[l2\ \beta2] -$$
$$2.\ EI2\ J2\ \beta2^5 \left(\frac{EI2\ \beta2^4}{\rho2}\right)^{1.} \mathrm{Cosh}[l2\ \beta2]\ \mathrm{Sin}[l2\ \beta2] + 1.\ EI2^2\ \beta2^6\ \mathrm{Sin}[l2\ \beta2]^2 +$$
$$1.\ J2\ m2\ \beta2^2 \left(\frac{EI2\ \beta2^4}{\rho2}\right)^{2.} \mathrm{Sin}[l2\ \beta2]^2 + 2.\ EI2\ m2\ \beta2^3 \left(\frac{EI2\ \beta2^4}{\rho2}\right)^{1.} \mathrm{Cos}[l2\ \beta2]\ \mathrm{Sinh}[l2\ \beta2] -$$
$$2.\ EI2\ J2\ \beta2^5 \left(\frac{EI2\ \beta2^4}{\rho2}\right)^{1.} \mathrm{Cos}[l2\ \beta2]\ \mathrm{Sinh}[l2\ \beta2] -$$
$$1.\ EI2^2\ \beta2^6\ \mathrm{Sinh}[l2\ \beta2]^2 - 1.\ J2\ m2\ \beta2^2 \left(\frac{EI2\ \beta2^4}{\rho2}\right)^{2.} \mathrm{Sinh}[l2\ \beta2]^2$$

In[∘]:=
```
simplifiedDeterminantbeam1 = FullSimplify[determinantbeam1]
simplifiedDeterminantbeam2 = FullSimplify[determinantbeam2]
```

Out[∘]=
$$J1\, m1\, \beta1^2 \left(\frac{EI1\, \beta1^4}{\rho1}\right)^{2.} \left(2. - 2. \operatorname{Cos}[l1\, \beta1] \operatorname{Cosh}[l1\, \beta1]\right) + \frac{1}{\rho1} EI1^2\, \beta1^6 \left(2. \rho1 + \operatorname{Cosh}[l1\, \beta1] \left(2. \rho1 \operatorname{Cos}[l1\, \beta1] + \beta1 \left(-2. m1 - 2. J1\, \beta1^2\right) \operatorname{Sin}[l1\, \beta1]\right) + \beta1 \left(2. m1 - 2. J1\, \beta1^2\right) \operatorname{Cos}[l1\, \beta1] \operatorname{Sinh}[l1\, \beta1]\right)$$

Out[∘]=
$$J2\, m2\, \beta2^2 \left(\frac{EI2\, \beta2^4}{\rho2}\right)^{2.} \left(2. - 2. \operatorname{Cos}[l2\, \beta2] \operatorname{Cosh}[l2\, \beta2]\right) + \frac{1}{\rho2} EI2^2\, \beta2^6 \left(2. \rho2 + \operatorname{Cosh}[l2\, \beta2] \left(2. \rho2 \operatorname{Cos}[l2\, \beta2] + \beta2 \left(-2. m2 - 2. J2\, \beta2^2\right) \operatorname{Sin}[l2\, \beta2]\right) + \beta2 \left(2. m2 - 2. J2\, \beta2^2\right) \operatorname{Cos}[l2\, \beta2] \operatorname{Sinh}[l2\, \beta2]\right)$$

In[∘]:=
```
simpleDeterminantbeam1 =
 simplifiedDeterminantbeam1 /. {J1 -> J1Val, m1 -> m1Val, m2 -> m2Val}
simpleDeterminantbeam2 =
 simplifiedDeterminantbeam2 /. {J2 -> J2Val, m1 -> m1Val, m2 -> m2Val}
```

Out[∘]=
$$4.09452 \times 10^{-7}\, \beta1^2 \left(\frac{EI1\, \beta1^4}{\rho1}\right)^{2.} \left(2. - 2. \operatorname{Cos}[l1\, \beta1] \operatorname{Cosh}[l1\, \beta1]\right) + \frac{1}{\rho1} EI1^2\, \beta1^6 \left(2. \rho1 + \operatorname{Cosh}[l1\, \beta1] \left(2. \rho1 \operatorname{Cos}[l1\, \beta1] + \beta1 \left(-0.1 - 0.0000163781\, \beta1^2\right) \operatorname{Sin}[l1\, \beta1]\right) + \beta1 \left(0.1 - 0.0000163781\, \beta1^2\right) \operatorname{Cos}[l1\, \beta1] \operatorname{Sinh}[l1\, \beta1]\right)$$

Out[∘]=
$$1.82112 \times 10^{-7}\, \beta2^2 \left(\frac{EI2\, \beta2^4}{\rho2}\right)^{2.} \left(2. - 2. \operatorname{Cos}[l2\, \beta2] \operatorname{Cosh}[l2\, \beta2]\right) + \frac{1}{\rho2} EI2^2\, \beta2^6 \left(2. \rho2 + \operatorname{Cosh}[l2\, \beta2] \left(2. \rho2 \operatorname{Cos}[l2\, \beta2] + \beta2 \left(-0.08 - 9.10558 \times 10^{-6}\, \beta2^2\right) \operatorname{Sin}[l2\, \beta2]\right) + \beta2 \left(0.08 - 9.10558 \times 10^{-6}\, \beta2^2\right) \operatorname{Cos}[l2\, \beta2] \operatorname{Sinh}[l2\, \beta2]\right)$$

In[ ]:=
```
paramValues = {EI1 -> EI1Val, EI2 -> EI2Val, l1 -> l1Val, l2 -> l2Val,
   J1 -> J1Val, J2 -> J2Val, ρ1 -> ρ1Val, ρ2 -> ρ2Val, m1 -> m1Val, m2 -> m2Val};

(*Substitute the numerical values into the determinant expression*)
simplifiedDeterminantbeam1 = Simplify[simpleDeterminantbeam1];
simplifiedDeterminantbeam2 = Simplify[simpleDeterminantbeam2];
numericalDeterminantbeam1 = simplifiedDeterminantbeam1 /. paramValues;
numericalDeterminantbeam2 = simplifiedDeterminantbeam2 /. paramValues;

(*Use NSolve to find solutions within the range*)
solutionsbeam1 = NSolve[{numericalDeterminantbeam1 == 0, 0 < β1 < 1000}, β1];
solutionsbeam2 = NSolve[{numericalDeterminantbeam2 == 0, 0 < β2 < 1000}, β2];

(*Extract unique solutions*)
uniqueSolutionsbeam1 = DeleteDuplicates[solutionsbeam1];
uniqueSolutionsbeam2 = DeleteDuplicates[solutionsbeam2];

β1Val = β1 /. uniqueSolutionsbeam1;
β2Val = β2 /. uniqueSolutionsbeam2;

β1sol = β1Val[[1]]
ωbeam1 = ω21 /. {β1 -> β1sol, EI1 -> EI1Val, EI2 -> EI2Val, l1 -> l1Val,
   l2 -> l2Val, J -> JVal, ρ1 -> ρ1Val, ρ2 -> ρ2Val, m1 -> m1Val, m2 -> m2Val}

β2sol = β2Val[[1]]
ωbeam2 = ω22 /. {β2 -> β2sol, EI1 -> EI1Val, EI2 -> EI2Val, l1 -> l1Val,
   l2 -> l2Val, J -> JVal, ρ1 -> ρ1Val, ρ2 -> ρ2Val, m1 -> m1Val, m2 -> m2Val}
```

Out[ ]= 7.12902

Out[ ]= 35.2394

Out[ ]= 7.96177

Out[ ]= 43.9529

In[ ]:=
```
k = 860;
masstot = MVal + m1Val + m2Val;
ω1 = (k / masstot) ^0.5
```

Out[ ]= 51.3618

In[ ]:=
```
ξ1 = 0.0073;
ξ2 = 0.017;
c1Val = ξ1 * 2 * masstot * ω1
c21Val = ξ2 * 2 * m1Val * ωbeam1
c22Val = ξ2 * 2 * m2Val * ωbeam2
```

Out[ ]= 0.244462

Out[ ]= 0.0599069

Out[ ]= 0.0597759

In[ ]:=
```
A1int = t1;
B1int = - (coefficientMatrixbeam1[[3, 1]] - coefficientMatrixbeam1[[3, 3]]) *
   t1 / (coefficientMatrixbeam1[[3, 2]] - coefficientMatrixbeam1[[3, 4]]);
C1int = -t1;
D1int = (coefficientMatrixbeam1[[3, 1]] - coefficientMatrixbeam1[[3, 3]]) *
   t1 / (coefficientMatrixbeam1[[3, 2]] - coefficientMatrixbeam1[[3, 4]]);

A2int = t2;
B2int = - (coefficientMatrixbeam2[[3, 1]] - coefficientMatrixbeam2[[3, 3]]) *
   t2 / (coefficientMatrixbeam2[[3, 2]] - coefficientMatrixbeam2[[3, 4]]);
C2int = -t2;
D2int = (coefficientMatrixbeam2[[3, 1]] - coefficientMatrixbeam2[[3, 3]]) *
   t2 / (coefficientMatrixbeam2[[3, 2]] - coefficientMatrixbeam2[[3, 4]]);

(*Display the relationships*)
{A1int, B1int, C1int, D1int}
{A2int, B2int, C2int, D2int}
```

Out[ ]=

$$\Big\{t1,\ \Big(t1\Big(1.\,J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Cos}[l1\,\beta1] - 1.\,J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Cosh}[l1\,\beta1] + 1.\,EI1\,\beta1^2\,\mathrm{Sin}[l1\,\beta1] + EI1\,\beta1^2\,\mathrm{Sinh}[l1\,\beta1]\Big)\Big) \Big/ \Big(-1.\,EI1\,\beta1^2\,\mathrm{Cos}[l1\,\beta1] - EI1\,\beta1^2\,\mathrm{Cosh}[l1\,\beta1] + J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Sin}[l1\,\beta1] + 1.\,J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Sinh}[l1\,\beta1]\Big),\ -t1,$$

$$\Big(t1\Big(-1.\,J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Cos}[l1\,\beta1] + 1.\,J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Cosh}[l1\,\beta1] - 1.\,EI1\,\beta1^2\,\mathrm{Sin}[l1\,\beta1] - EI1\,\beta1^2\,\mathrm{Sinh}[l1\,\beta1]\Big)\Big) \Big/ \Big(-1.\,EI1\,\beta1^2\,\mathrm{Cos}[l1\,\beta1] - EI1\,\beta1^2\,\mathrm{Cosh}[l1\,\beta1] + J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Sin}[l1\,\beta1] + 1.\,J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Sinh}[l1\,\beta1]\Big)\Big\}$$

Out[•]=

$$\left\{t2,\ \left(t2\left(1.\,J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Cos}[l2\,\beta2]-1.\,J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Cosh}[l2\,\beta2]+1.\,EI2\,\beta2^2\,\mathrm{Sin}[l2\,\beta2]+EI2\,\beta2^2\,\mathrm{Sinh}[l2\,\beta2]\right)\right)\Big/\left(-1.\,EI2\,\beta2^2\,\mathrm{Cos}[l2\,\beta2]-EI2\,\beta2^2\,\mathrm{Cosh}[l2\,\beta2]+J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Sin}[l2\,\beta2]+1.\,J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Sinh}[l2\,\beta2]\right),\ -t2,\ \left(t2\left(-1.\,J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Cos}[l2\,\beta2]+1.\,J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Cosh}[l2\,\beta2]-1.\,EI2\,\beta2^2\,\mathrm{Sin}[l2\,\beta2]-EI2\,\beta2^2\,\mathrm{Sinh}[l2\,\beta2]\right)\right)\Big/\left(-1.\,EI2\,\beta2^2\,\mathrm{Cos}[l2\,\beta2]-EI2\,\beta2^2\,\mathrm{Cosh}[l2\,\beta2]+J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Sin}[l2\,\beta2]+1.\,J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Sinh}[l2\,\beta2]\right)\right\}$$

In[∘]:=
```
(*Substitute Ai,Bi,Ci,Di with At,Bt,Ct,Dt*)
ϕ1n = ϕ1 /. {A1i -> A1int, B1i -> B1int, C1i -> C1int, D1i -> D1int};
ϕ1n
ϕ2n = ϕ2 /. {A2i -> A2int, B2i -> B2int, C2i -> C2int, D2i -> D2int};
ϕ2n
```

Out[∘]=

$$t1\,\mathrm{Sin}[s\,\beta1] + \left(t1\,\mathrm{Cosh}[s\,\beta1]\left(-1.\,J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Cos}[l1\,\beta1] + 1.\,J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Cosh}[l1\,\beta1] - 1.\,EI1\,\beta1^2\,\mathrm{Sin}[l1\,\beta1] - EI1\,\beta1^2\,\mathrm{Sinh}[l1\,\beta1]\right)\right) \Big/ \left(-1.\,EI1\,\beta1^2\,\mathrm{Cos}[l1\,\beta1] - EI1\,\beta1^2\,\mathrm{Cosh}[l1\,\beta1] + J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Sin}[l1\,\beta1] + 1.\,J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Sinh}[l1\,\beta1]\right) + \left(t1\,\mathrm{Cos}[s\,\beta1]\left(1.\,J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Cos}[l1\,\beta1] - 1.\,J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Cosh}[l1\,\beta1] + 1.\,EI1\,\beta1^2\,\mathrm{Sin}[l1\,\beta1] + EI1\,\beta1^2\,\mathrm{Sinh}[l1\,\beta1]\right)\right) \Big/ \left(-1.\,EI1\,\beta1^2\,\mathrm{Cos}[l1\,\beta1] - EI1\,\beta1^2\,\mathrm{Cosh}[l1\,\beta1] + J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Sin}[l1\,\beta1] + 1.\,J1\,\beta1\left(\frac{EI1\,\beta1^4}{\rho1}\right)^{1.}\mathrm{Sinh}[l1\,\beta1]\right) - t1\,\mathrm{Sinh}[s\,\beta1]$$

Out[∘]=

$$t2\,\mathrm{Sin}[s\,\beta2] + \left(t2\,\mathrm{Cosh}[s\,\beta2]\left(-1.\,J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Cos}[l2\,\beta2] + 1.\,J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Cosh}[l2\,\beta2] - 1.\,EI2\,\beta2^2\,\mathrm{Sin}[l2\,\beta2] - EI2\,\beta2^2\,\mathrm{Sinh}[l2\,\beta2]\right)\right) \Big/ \left(-1.\,EI2\,\beta2^2\,\mathrm{Cos}[l2\,\beta2] - EI2\,\beta2^2\,\mathrm{Cosh}[l2\,\beta2] + J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Sin}[l2\,\beta2] + 1.\,J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Sinh}[l2\,\beta2]\right) + \left(t2\,\mathrm{Cos}[s\,\beta2]\left(1.\,J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Cos}[l2\,\beta2] - 1.\,J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Cosh}[l2\,\beta2] + 1.\,EI2\,\beta2^2\,\mathrm{Sin}[l2\,\beta2] + EI2\,\beta2^2\,\mathrm{Sinh}[l2\,\beta2]\right)\right) \Big/ \left(-1.\,EI2\,\beta2^2\,\mathrm{Cos}[l2\,\beta2] - EI2\,\beta2^2\,\mathrm{Cosh}[l2\,\beta2] + J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Sin}[l2\,\beta2] + 1.\,J2\,\beta2\left(\frac{EI2\,\beta2^4}{\rho2}\right)^{1.}\mathrm{Sinh}[l2\,\beta2]\right) - t2\,\mathrm{Sinh}[s\,\beta2]$$

In[∘]:=
```
ϕ1nD2 = D[ϕ1n, {s, 2}];
ϕ2nD2 = D[ϕ2n, {s, 2}];
paramValues =
  {EI1 -> EI1Val, EI2 -> EI2Val, l1 -> l1Val, l2 -> l2Val, J1 -> J1Val, J2 -> J2Val,
   ρ1 -> ρ1Val, ρ2 -> ρ2Val, m1 -> m1Val, m2 -> m2Val, β1 -> β1sol, β2 -> β2sol};

OrthogonalEqnbeam1 = EI1 * Integrate[ϕ1nD2^2, {s, 0, l1}] == ωbeam1^2;
OrthogonalEqnbeam2 = EI2 * Integrate[ϕ2nD2^2, {s, 0, l2}] == ωbeam2^2;

OrthoBeam1 = Simplify[OrthogonalEqnbeam1 /. paramValues];
OrthoBeam2 = Simplify[OrthogonalEqnbeam2 /. paramValues];

solutiont1 = NSolve[OrthogonalEqnbeam1 /. paramValues, t1];
solutiont2 = NSolve[OrthogonalEqnbeam2 /. paramValues, t2];

Chop[solutiont1]
Chop[solutiont2]
```

Out[∘]= {{t1 → -39.8519}, {t1 → 39.8519}}

Out[∘]= {{t2 → -40.1253}, {t2 → 40.1253}}

In[∘]:=
```
sol1 = Chop[solutiont1[[2, 1, 2]]]
sol2 = Chop[solutiont2[[2, 1, 2]]]
```

Out[∘]= 39.8519

Out[∘]= 40.1253

In[ ]:=

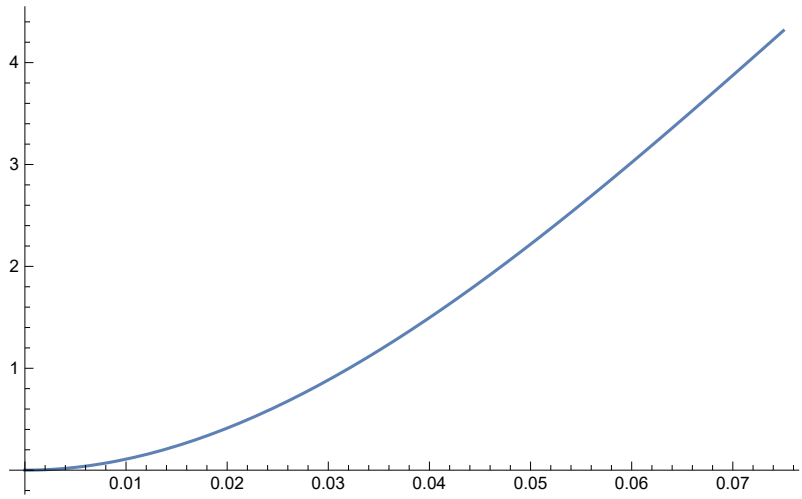
```
t1Value = t1 /. t1 -> sol1;
t2Value = t2 /. t2 -> sol2;
A1Val = A1int /. t1 -> t1Value;
B1Val = B1int /. t1 -> t1Value;
C1Val = C1int /. t1 -> t1Value;
D1Val = D1int /. t1 -> t1Value;

A2Val = A2int /. t2 -> t2Value;
B2Val = B2int /. t2 -> t2Value;
C2Val = C2int /. t2 -> t2Value;
D2Val = D2int /. t2 -> t2Value;
paramValues =
  {EI1 -> EI1Val, EI2 -> EI2Val, l1 -> l1Val, l2 -> l2Val, J1 -> J1Val, J2 -> J2Val,
   ρ1 -> ρ1Val, ρ2 -> ρ2Val, m1 -> m1Val, m2 -> m2Val, β1 -> β1sol, β2 -> β2sol};

ϕ1f = ϕ1n /. {A1int -> A1Val, B1int -> B1Val, C1int -> C1Val, D1int -> D1Val};
ϕ2f = ϕ2n /. {A2int -> A2Val, B1int -> B2Val, C2int -> C2Val, D2int -> D2Val};

ϕ1modeshape = ϕ1f /. paramValues;
ϕ2modeshape = ϕ2f /. paramValues;

ϕ1num = Simplify[ϕ1modeshape];
ϕ2num = Simplify[ϕ2modeshape];

ϕ1num
ϕ2num
```

Out[ ]= $-22.186\,\mathrm{Cos}[7.12902\,s] + 22.186\,\mathrm{Cosh}[7.12902\,s] + 39.8519\,\mathrm{Sin}[7.12902\,s] - 39.8519\,\mathrm{Sinh}[7.12902\,s]$

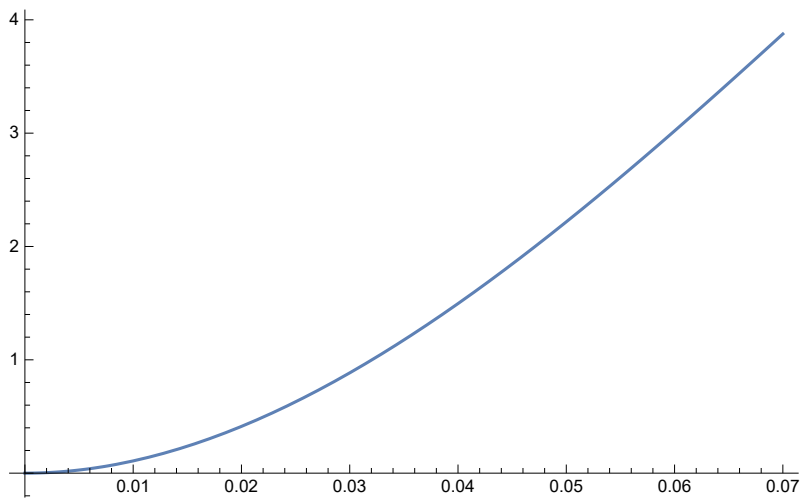
Out[ ]= $-23.0698\,\mathrm{Cos}[7.96177\,s] + 23.0698\,\mathrm{Cosh}[7.96177\,s] + 40.1253\,\mathrm{Sin}[7.96177\,s] - 40.1253\,\mathrm{Sinh}[7.96177\,s]$

In[ ]:=
```
s1Range = {s, 0, l1Val};(*Example range from 0 to l_val*)
(*Plot the function PhiFinal*)
Plot[Evaluate[ϕ1num], s1Range]
s2Range = {s, 0, l2Val};(*Example range from 0 to l_val*)
(*Plot the function PhiFinal*)
Plot[Evaluate[ϕ1num], s2Range]
```

Out[ ]=



Out[ ]=



In[∘]:=
```
paramValues = {EI1 -> EI1Val, EI2 -> EI2Val, l1 -> l1Val,
   l2 -> l2Val, J1 -> J1Val, J2 -> J2Val, ρ1 -> ρ1Val, ρ2 -> ρ2Val, m1 -> m1Val,
   m2 -> m2Val, c1 -> c1Val, c21 -> c21Val, c22 -> c22Val, M -> MVal};
m1bar = Integrate[ρ1, {s, 0, l1}] /. paramValues;
m2bar = Integrate[ρ2, {s, 0, l2}] /. paramValues;
masstot = (m1 + m2 + M) /. paramValues;
η11 = Integrate[c21 * ϕ1num^2, {s, 0, l1}] /. paramValues
η12 =
 Integrate[EI1 * (ϕ1num * D[ϕ1num, s]^2 * D[ϕ1num, {s, 4}] + 4 * ϕ1num * D[ϕ1num, s] * D[ϕ1num,
        {s, 2}] * D[ϕ1num, {s, 3}] + ϕ1num * D[ϕ1num, {s, 2}]^3), {s, 0, l1}] /. paramValues
η13 = m1 * (Integrate[D[ϕ1num, s]^2, {s, 0, l1}])^2 /. paramValues
η14 = m1 * Integrate[D[ϕ1num, s]^2, {s, 0, l1}] /. paramValues

η21 = Integrate[c22 * ϕ2num^2, {s, 0, l2}] /. paramValues
η22 =
 Integrate[EI2 * (ϕ2num * D[ϕ2num, s]^2 * D[ϕ2num, {s, 4}] + 4 * ϕ2num * D[ϕ2num, s] * D[ϕ2num,
        {s, 2}] * D[ϕ2num, {s, 3}] + ϕ2num * D[ϕ2num, {s, 2}]^3), {s, 0, l2}] /. paramValues
η23 = m2 * (Integrate[D[ϕ2num, s]^2, {s, 0, l2}])^2 /. paramValues
η24 = m2 * Integrate[D[ϕ2num, s]^2, {s, 0, l2}] /. paramValues

X0 = 0.01;
ε1 = η14 * X0 /. paramValues;
ε2 = η24 * X0 /. paramValues;
R1 = masstot / m1 /. paramValues;
R2 = masstot / m2 /. paramValues;
μ21 = η11 / (2 * ε1) /. paramValues;
μ22 = η21 / (2 * ε2) /. paramValues;
μ1 = c1 / (2 * ε1 * masstot) /. paramValues;
μ = c1 / (MVal) /. paramValues;
δ1 = masstot * η12 / η14^2 /. paramValues;
δ2 = masstot * η22 / η24^2 /. paramValues;
F0 = 100;
F = F0 * masstot * X0;
```

Out[∘]= 0.019513

Out[∘]= -5.11675×10^{6}

Out[∘]= 4475.37

Out[∘]= 14.9589

Out[∘]= 0.0230422

Out[∘]= -1.14817×10^{7}

Out[∘]= 6566.75

Out[∘]= 16.2071

In[ ]:=
```
α = ω1 / ωbeam1
β = ω1 / ωbeam2
r = 2 / α
Ω = r * ω1
(*Find steady-state amplitudes*)
(*steadyStateInterval={1800,1900};
amplitudeX=FindMaximum[Evaluate[x[t]/. solution],{t,1800,1900}][[1]];
amplitudeQ1=FindMaximum[Evaluate[q1[t]/. solution],{t,1800,1900}][[1]];
amplitudeQ2=FindMaximum[Evaluate[q2[t]/. solution],{t,1800,1900}][[1]];
{amplitudeX,amplitudeQ1,amplitudeQ2}*)
```

Out[ ]= 1.45751

Out[ ]= 1.16857

Out[ ]= 1.3722

Out[ ]= 70.4787

In[ ]:=
```
eq21 = q1''[t] + 2 * ε1 * μ21 * q1'[t] + ωbeam1^2 * q1[t] + δ1 * ε1^2 * q1[t]^3 +
   R1 * ε1^2 * q1[t] * (q1'[t]^2 + q1[t] * q1''[t]) - ε1 * q1[t] * x''[t] == 0
eq22 = q2''[t] + 2 * ε2 * μ22 * q2'[t] + ωbeam2^2 * q2[t] + δ2 * ε2^2 * q2[t]^3 +
   R2 * ε2^2 * q2[t] * (q2'[t]^2 + q2[t] * q2''[t]) - ε2 * q2[t] * x''[t] == 0
eq1 = x''[t] + 2 * ε1 * μ1 * x'[t] + ω1^2 * x[t] - ε1 * (q1'[t]^2 + q1[t] * q1''[t]) -
   ε2 * (q2'[t]^2 + q2[t] * q2''[t]) == F0 * Cos[Ω * t]
(*Set initial conditions*)
initialConditions =
  {q1[0] == 10^-10, q1'[0] == 0, q2[0] == 10^-10, q2'[0] == 0, x[0] == 0, x'[0] == 0};

(*Solve the system numerically*)
solution = NDSolve[{eq1, eq21, eq22, initialConditions}, {x, q1, q2}, {t, 0, 500}];
```

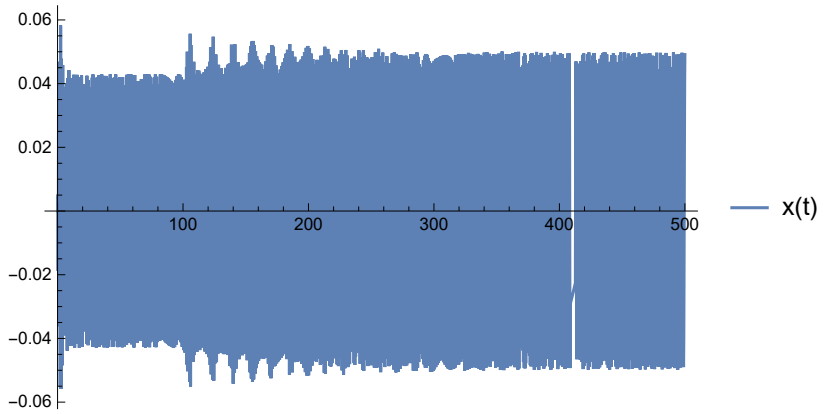
Out[ ]= $1241.81\, q1[t] - 166.806\, q1[t]^3 + 0.019513\, q1'[t] + q1''[t] + 0.145897\, q1[t]\, (q1'[t]^2 + q1[t]\, q1''[t]) - 0.149589\, q1[t]\, x''[t] == 0$

Out[ ]= $1931.86\, q2[t] - 374.304\, q2[t]^3 + 0.0230422\, q2'[t] + q2''[t] + 0.214076\, q2[t]\, (q2'[t]^2 + q2[t]\, q2''[t]) - 0.162071\, q2[t]\, x''[t] == 0$

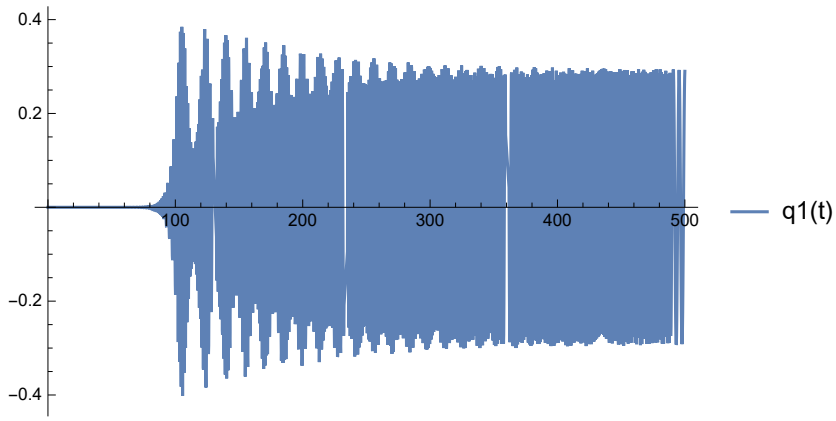
Out[ ]= $2638.04\, x[t] + 0.749883\, x'[t] - 0.149589\, (q1'[t]^2 + q1[t]\, q1''[t]) - 0.162071\, (q2'[t]^2 + q2[t]\, q2''[t]) + x''[t] == 100\, Cos[70.4787\, t]$

In[ ]:=
```
(*Plot the solutions*)
Plot[Evaluate[{x[t]} /. solution], {t, 0, 500}, PlotLegends -> {"x(t)"}, PlotRange -> All]
Plot[Evaluate[{q1[t]} /. solution],
 {t, 0, 500}, PlotLegends -> {"q1(t)"}, PlotRange -> All]
Plot[Evaluate[{q2[t]} /. solution], {t, 0, 500}, PlotLegends -> {"q2(t)"}, PlotRange -> All]
```

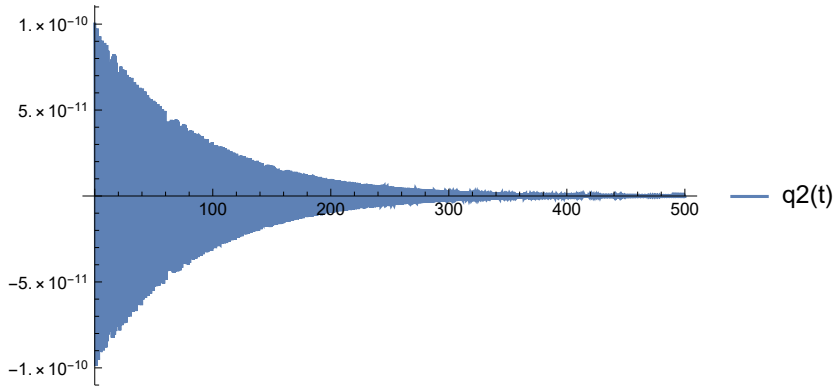
Out[ ]=



Out[ ]=



Out[ ]=



In[ ]:=
```
t = 0.1;
x[t] = 0.1;
x'[t] = 0.1;
q1[t] = 0.1;
q1'[t] = 0.1;
q2[t] = 0.1;
q2'[t] = 0.1;
Ω = 0.1;
q1''[t] = -126.3578;
q2''[t] = -195.1371;
eq1 = x''[t] + 2 * ε1 * μ1 * x'[t] + ω1^2 * x[t] -
   ε1 * (q1'[t]^2 + q1[t] * q1''[t]) - ε2 * (q2'[t]^2 + q2[t] * q2''[t]) == F0 * Cos[Ω * t]
```

Out[ ]=
```
268.928 + x''[0.1] == 99.995
```